

AI CRM Lead Automation

A safe, testable workflow that converts inbound messages into structured CRM records, priorities, follow-up dates, and spreadsheet-safe exports.

DELIVERY	ANALYSIS	QUALITY	BOUNDARY
Local dashboard + API	Offline or optional OpenAI	39 automated tests	Reference app, not SaaS

The business problem

Small teams receive inquiries through forms, inboxes, chat, and referrals. Manual interpretation and copy-paste work delay responses, lose context, and make it hard to see which lead needs attention first.

The implemented solution

The local application validates each intake, extracts contact details, classifies the request, calculates a transparent Hot/Warm/Cold score, schedules follow-up, checks duplicate emails, and writes an audit event. A responsive dashboard, JSON endpoints, and CSV export make the result immediately reviewable.

Two analysis modes, one contract

DEFAULT: DETERMINISTIC	OPTIONAL: OPENAI
Offline, reproducible, zero API cost. Keyword and rule-based categories feed the same transparent score and follow-up logic.	Responses API with strict Structured Outputs, store: false, bounded retries, and a second local schema validation pass.

Workflow

1 Bounded JSON intake Media type, byte count, object shape, source, and message length are validated.	2 Contact + classification Deterministic rules or validated model output become bounded fields.	3 Atomic CRM write Lead, duplicate state, and event log are persisted under one process lock.
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Built for credible review

The portfolio value is not a claim of a live CRM deployment. It is an inspectable reference architecture with explicit privacy, security, and operating boundaries.

Reliable persistence	Safer interfaces
- Thread-safe read/modify/write operations - Atomic tempfile + replace writes - Corrupt JSON shape detection - Runtime data excluded from Git	- Generic server errors - HTML escaping and allowlisted classes - Formula-safe CSV cells - No-cache and browser security headers

Privacy by default

- Raw inquiry text is not retained unless STORE_RAW_MESSAGE=1 is deliberately enabled.
- API keys, provider bodies, and lead text are excluded from logs.
- OpenAI mode sends only the normalized inquiry needed for the requested classification and sets store: false.

Verification evidence

39 isolated tests	3 Python versions in CI	2 reproducible PDFs
Analysis, extraction, provider, storage, concurrency, CSV, HTML, and local HTTP behavior.	Python 3.11, 3.12, and 3.13 run the same standard-library suite.	Build, page count, metadata, and required text are checked in CI.

Client adaptation path

After requirements discovery, the same validated record can be connected to an authenticated form or inbox and written through an official CRM API. Production delivery would add identity and access controls, TLS, rate limits, a managed transactional database, retention/deletion automation, monitoring, backups, and reconciliation.

Honest scope

This repository demonstrates the engineering pattern. It does not claim active Gmail, HubSpot, Airtable, Google Sheets, Pipedrive, GoHighLevel, Zapier, Make, or n8n integrations. A paid live OpenAI smoke test also requires the project owner's API key; all other behavior is locally and continuously verified.